

Real-time scenario-based interview questions for Azure Cosmos DB Administration, Performance Tuning & Optimization:

Covering from **basic setup to extreme performance tuning, multi-region replication, backup/recovery, and security.**

Azure Cosmos DB – 500 Real-Time Scenario-Based Interview Questions

Basic Setup & Configuration (1–100)

1. How do you create a Cosmos DB account using the Azure Portal?
2. How do you select between SQL API, MongoDB API, Cassandra API, Gremlin API, and Table API?
3. How do you choose between single-region and multi-region Cosmos DB accounts?
4. How do you configure a database and container in Cosmos DB?
5. How do you define a partition key for a container?
6. How do you configure throughput as provisioned RU/s vs. serverless?
7. How do you enable multi-master writes in Cosmos DB?
8. How do you enable geo-redundancy for a database?
9. How do you define TTL (Time to Live) for automatic item expiration?
10. How do you enable consistency levels (Strong, Bounded Staleness, Session, Consistent Prefix, Eventual)?
11. How do you monitor throughput usage in real-time?
12. How do you configure auto-scale throughput?
13. How do you add a new container to an existing database?
14. How do you remove or delete a container safely?
15. How do you scale RU/s dynamically to handle workload spikes?
16. How do you monitor throttled requests due to RU limits?
17. How do you detect and avoid hot partitions?
18. How do you optimize partition key selection for uniform data distribution?
19. How do you configure indexing policies for efficient queries?
20. How do you implement schema design for time-series data?
21. How do you enable analytical store for Cosmos DB for analytical queries?
22. How do you monitor storage consumption across containers?
23. How do you configure unique key constraints on a container?
24. How do you enable client-side encryption for sensitive data?
25. How do you integrate Cosmos DB with Azure Active Directory for access control?
26. How do you define stored procedures for server-side logic?
27. How do you define triggers and user-defined functions in Cosmos DB?
28. How do you monitor request charge (RU) per query or operation?
29. How do you optimize queries to minimize RU consumption?
30. How do you detect queries causing excessive RU consumption?
31. How do you implement batch operations efficiently in Cosmos DB?
32. How do you detect operations causing throttling during peak workloads?
33. How do you design containers for multi-tenant applications?
34. How do you monitor item-level performance metrics?
35. How do you detect slow queries affecting SLA compliance?
36. How do you optimize query filters and projections for better RU utilization?

37. How do you monitor change feed for data processing?
38. How do you implement safe cross-partition queries?
39. How do you configure partitioned vs. non-partitioned containers?
40. How do you handle large item sizes exceeding 2 MB?
41. How do you monitor and manage request units (RU/s) per operation?
42. How do you detect cross-partition queries causing high RU consumption?
43. How do you configure backup policies for continuous and periodic backups?
44. How do you restore data from a backup in Cosmos DB?
45. How do you implement disaster recovery across multiple regions?
46. How do you monitor geo-replication status?
47. How do you detect replication conflicts in multi-master setups?
48. How do you optimize read and write operations using session tokens?
49. How do you monitor container-level metrics in Azure Monitor?
50. How do you implement change feed processors efficiently?
51. How do you configure analytical store for Synapse integration?
52. How do you monitor latency trends for queries and writes?
53. How do you detect queries causing hotspots in partitions?
54. How do you optimize indexing paths to reduce RU cost?
55. How do you implement TTL for archiving old data?
56. How do you monitor request charge per stored procedure execution?
57. How do you optimize stored procedures for high concurrency?
58. How do you detect and handle rate-limiting errors (HTTP 429)?
59. How do you monitor read/write latency across regions?
60. How do you implement caching to reduce RU consumption?
61. How do you define container throughput for heavy write workloads?
62. How do you monitor storage growth trends for capacity planning?
63. How do you detect cross-region replication lag?
64. How do you optimize queries using continuation tokens for pagination?
65. How do you monitor analytical store query performance?
66. How do you detect slow change feed processing?
67. How do you implement partition key strategies for uniform data distribution?
68. How do you monitor request charge spikes during peak load?
69. How do you detect inefficient queries using Azure Monitor?
70. How do you implement automated alerts for high RU consumption?
71. How do you configure unique keys to prevent duplicate data insertion?
72. How do you optimize queries for attributes with high cardinality?
73. How do you monitor multi-region write availability?
74. How do you detect and resolve partition key hotspots?
75. How do you optimize indexing policies for selective attributes?
76. How do you monitor change feed latency for SLA compliance?
77. How do you detect high-latency queries caused by cross-partition operations?
78. How do you optimize batch inserts to reduce throttling?
79. How do you monitor container throughput and storage usage in real-time?

80. How do you implement retries for transient failures efficiently?
81. How do you configure analytical store retention policies?
82. How do you monitor container-level RU/s utilization trends?
83. How do you detect multi-region replication conflicts automatically?
84. How do you optimize queries for large datasets using continuation tokens?
85. How do you monitor request charge for complex queries?
86. How do you detect inefficient query projections?
87. How do you optimize queries for frequent updates and deletes?
88. How do you implement server-side logic to reduce client-side round-trips?
89. How do you monitor container-level latency under heavy workloads?
90. How do you detect queries causing excessive cross-partition traffic?
91. How do you implement TTL-based cleanup for time-series data?
92. How do you monitor throughput utilization across multiple containers?
93. How do you detect throttling caused by multi-master writes?
94. How do you optimize queries using selective index paths?
95. How do you monitor change feed processing latency and errors?
96. How do you detect queries causing hot partitions in high-volume workloads?
97. How do you optimize batch operations to avoid RU spikes?
98. How do you monitor geo-replication synchronization status?
99. How do you detect and resolve read/write conflicts in multi-region setup?
100. How do you configure provisioned throughput for extreme high-traffic scenarios?

Query Performance & Index Optimization (101–180)

101. How do you optimize queries to minimize cross-partition operations?
102. How do you detect slow queries using Azure Monitor metrics?
103. How do you optimize indexing paths for selective queries?
104. How do you implement continuation tokens to handle large query results?
105. How do you monitor RU consumption per query for cost optimization?
106. How do you detect queries causing excessive latency under peak load?
107. How do you optimize queries for frequently filtered attributes?
108. How do you detect cross-partition queries causing throttling?
109. How do you monitor query efficiency across multiple containers?
110. How do you implement efficient projections to reduce RU cost?
111. How do you detect queries performing full container scans unnecessarily?
112. How do you optimize queries for high-cardinality attributes?
113. How do you monitor query latency trends for performance tuning?
114. How do you implement parameterized queries to improve RU efficiency?
115. How do you detect queries that bypass indexes causing performance degradation?
116. How do you optimize query filters for multi-attribute conditions?
117. How do you monitor request charge per query execution?
118. How do you detect queries causing network latency issues?
119. How do you optimize queries for time-range and sorting operations?
120. How do you implement server-side filtering to reduce client-side processing?

121. How do you detect queries that are not using partition keys efficiently?
122. How do you optimize batch read operations for large datasets?
123. How do you monitor slow query patterns using Azure Monitor logs?
124. How do you detect queries causing uneven partition usage?
125. How do you optimize queries for multi-tenant applications?
126. How do you monitor RU spikes during analytical queries?
127. How do you detect queries causing high CPU utilization in client applications?
128. How do you optimize queries for large text or binary attributes?
129. How do you monitor request latency in multi-region deployments?
130. How do you detect inefficient query projections affecting RU consumption?
131. How do you optimize cross-partition queries for multi-region setups?
132. How do you detect and reduce throttled queries in high-concurrency scenarios?
133. How do you monitor GSI and indexing impact on query performance?
134. How do you optimize query patterns to reduce RU cost?
135. How do you detect queries causing hot partitions in Cosmos DB containers?
136. How do you implement efficient pagination for large result sets?
137. How do you monitor cross-partition query latency?
138. How do you detect inefficient conditional queries affecting performance?
139. How do you optimize queries for frequent inserts and updates?
140. How do you monitor analytical store queries for resource consumption?
141. How do you detect queries causing excessive cross-region traffic?
142. How do you optimize queries using partition key and sort key effectively?
143. How do you monitor batch query execution times?
144. How do you detect queries causing high RU charges in peak hours?
145. How do you optimize queries for selective attributes in large containers?
146. How do you monitor multi-region write latency for performance tuning?
147. How do you detect queries bypassing efficient indexing paths?
148. How do you optimize queries for hierarchical or nested JSON structures?
149. How do you monitor change feed query performance?
150. How do you detect and resolve queries causing excessive RU spikes?
151. How do you implement query retry logic for rate-limited operations?
152. How do you monitor query patterns affecting hot partitions?
153. How do you detect inefficient server-side operations impacting throughput?
154. How do you optimize queries for frequently updated properties?
155. How do you monitor query execution cost for SLA compliance?
156. How do you detect queries causing high network latency?
157. How do you optimize queries for mixed read/write workloads?
158. How do you monitor query performance for multi-tenant scenarios?
159. How do you detect queries causing cross-partition write conflicts?
160. How do you optimize queries for high-throughput analytical processing?
161. How do you monitor query efficiency for time-series workloads?
162. How do you detect queries causing request throttling?
163. How do you optimize query execution plans for large JSON documents?

164. How do you monitor cross-partition query distribution?
165. How do you detect queries causing storage hotspots?
166. How do you optimize queries for high-frequency updates?
167. How do you monitor RU consumption trends over time?
168. How do you detect queries performing inefficient filter operations?
169. How do you optimize queries for multi-attribute range conditions?
170. How do you monitor queries causing latency spikes in production?
171. How do you detect queries bypassing indexing paths?
172. How do you optimize queries for global distribution scenarios?
173. How do you monitor query throughput during peak hours?
174. How do you detect cross-region queries causing high latency?
175. How do you optimize queries for container-level throughput efficiency?
176. How do you monitor query execution patterns for cost optimization?
177. How do you detect queries causing excessive server-side processing?
178. How do you optimize queries for large hierarchical datasets?
179. How do you monitor continuation token usage for large result sets?
180. How do you detect queries causing uneven RU utilization?

Multi-Region Replication & Global Distribution (181–250)

181. How do you configure multi-region write support for Cosmos DB?
182. How do you monitor replication latency across regions?
183. How do you detect conflicts in multi-master writes?
184. How do you implement conflict resolution policies (last-write-wins vs custom)?
185. How do you optimize read/write performance in a multi-region setup?
186. How do you monitor failover status in multi-region accounts?
187. How do you implement global distribution for low-latency reads?
188. How do you detect replication failures in near real-time?
189. How do you optimize partition key selection for global distribution?
190. How do you monitor cross-region request latencies using Azure Monitor?
191. How do you detect hot partitions affecting global writes?
192. How do you optimize read operations to use nearest region endpoints?
193. How do you monitor multi-region throughput usage trends?
194. How do you detect inconsistent data between regions?
195. How do you implement automated alerts for replication lag?
196. How do you optimize writes to avoid cross-region throttling?
197. How do you monitor global table consistency levels?
198. How do you detect conflicts in multi-region analytical workloads?
199. How do you implement disaster recovery using geo-redundant writes?
200. How do you monitor endpoint health for multi-region Cosmos DB accounts?
201. How do you detect replication bottlenecks under peak load?
202. How do you optimize GSI usage for multi-region scenarios?
203. How do you implement failover testing without downtime?
204. How do you monitor replication latency for SLA compliance?

205. How do you detect cross-region write conflicts automatically?
206. How do you optimize partition distribution for global workloads?
207. How do you monitor throughput distribution across regions?
208. How do you detect queries causing cross-region network congestion?
209. How do you optimize analytical queries on multi-region replicated data?
210. How do you monitor global write latency for high-traffic containers?
211. How do you detect inconsistent reads in eventually consistent regions?
212. How do you optimize cross-region data transfers to reduce RU consumption?
213. How do you monitor endpoint-specific latency in multi-region accounts?
214. How do you detect high-latency operations caused by multi-region replication?
215. How do you optimize queries for near real-time analytics across regions?
216. How do you monitor multi-region conflict resolution events?
217. How do you detect write hotspots in global containers?
218. How do you implement automated conflict resolution scripts?
219. How do you optimize container throughput for global distribution?
220. How do you monitor replication success rates for compliance?
221. How do you detect replication lag during high-concurrency writes?
222. How do you optimize partition key selection for multi-master writes?
223. How do you monitor cross-region latency for analytical store queries?
224. How do you detect and mitigate multi-region throttling issues?
225. How do you optimize queries for cross-region reads?
226. How do you monitor replication consistency using Azure Monitor metrics?
227. How do you detect conflicting writes in high-volume global workloads?
228. How do you optimize global distribution for cost efficiency?
229. How do you monitor throughput spikes across regions?
230. How do you detect uneven data distribution across global partitions?
231. How do you optimize analytical queries for multi-region replicated containers?
232. How do you monitor failover readiness for disaster recovery?
233. How do you detect performance degradation in multi-master writes?
234. How do you optimize change feed processing across regions?
235. How do you monitor replication lag for SLA reporting?
236. How do you detect and resolve cross-region data inconsistencies?
237. How do you optimize queries to reduce cross-region latency?
238. How do you monitor global write distribution patterns?
239. How do you detect inefficient multi-region GSI usage?
240. How do you optimize container throughput for low-latency global reads?
241. How do you monitor replication errors during peak load?
242. How do you detect hot partitions affecting global writes?
243. How do you implement automated alerts for multi-region issues?
244. How do you optimize query performance in a globally distributed database?
245. How do you monitor endpoint-specific request charge for cross-region operations?
246. How do you detect queries causing excessive RU consumption globally?
247. How do you optimize partition key design to avoid throttling in multi-region writes?

- 248. How do you monitor multi-region write availability for high-traffic workloads?
- 249. How do you detect replication bottlenecks affecting analytics?
- 250. How do you implement schema changes across global regions safely?

Backup, PITR & Recovery (251–320)

- 251. How do you enable continuous backup in Cosmos DB?
- 252. How do you restore data from a specific point-in-time?
- 253. How do you monitor backup job status and failures?
- 254. How do you restore data to a new container without affecting production?
- 255. How do you validate restored data integrity?
- 256. How do you schedule periodic backups without impacting performance?
- 257. How do you restore data across regions for disaster recovery?
- 258. How do you detect backup failures and automate retries?
- 259. How do you monitor storage consumption for backups?
- 260. How do you implement long-term retention policies for compliance?
- 261. How do you perform PITR restore for large containers efficiently?
- 262. How do you monitor incremental backup performance?
- 263. How do you detect failed restores automatically?
- 264. How do you restore only selected items or partitions?
- 265. How do you optimize restore performance for high-volume workloads?
- 266. How do you implement automated verification of restored data?
- 267. How do you monitor backup duration trends for performance planning?
- 268. How do you restore encrypted data securely?
- 269. How do you optimize backup frequency to balance cost and protection?
- 270. How do you implement cross-account backup and restore?
- 271. How do you restore multi-region containers with minimal downtime?
- 272. How do you monitor PITR restore metrics in Azure Monitor?
- 273. How do you detect anomalies in backup job duration?
- 274. How do you implement automated testing of backup restores?
- 275. How do you monitor backup job completion across multiple accounts?
- 276. How do you detect failed incremental backups?
- 277. How do you optimize restore for containers with multiple GSIs?
- 278. How do you monitor restore throttling to avoid RU spikes?
- 279. How do you restore tables to a different subscription or tenant?
- 280. How do you implement automated alerts for failed restores?
- 281. How do you detect partial restore failures in large containers?
- 282. How do you optimize restore throughput using parallel operations?
- 283. How do you restore containers while ongoing writes are happening?
- 284. How do you monitor restore success rates for compliance reporting?
- 285. How do you detect inconsistencies after a PITR restore?
- 286. How do you optimize backup storage costs without losing recovery points?
- 287. How do you monitor backup job performance for SLA adherence?
- 288. How do you detect restore errors caused by schema mismatches?

289. How do you implement automated cross-region restore testing?
290. How do you monitor container-level backup consumption trends?
291. How do you detect and mitigate backup-induced throttling?
292. How do you optimize restore strategies for containers with high write throughput?
293. How do you monitor PITR for multi-region containers?
294. How do you detect restore bottlenecks under peak load?
295. How do you implement selective item restore for business-critical data?
296. How do you monitor restore latency in multi-region scenarios?
297. How do you detect backup failures affecting global replication?
298. How do you optimize restore operations for containers with large payloads?
299. How do you implement automated verification of multi-region restores?
300. How do you monitor backup job success trends over time?
301. How do you detect data loss risks due to incomplete backups?
302. How do you optimize restore for containers with heavy analytical usage?
303. How do you monitor incremental backups for high-traffic containers?
304. How do you detect failed restore attempts due to insufficient permissions?
305. How do you implement automated backup rotation for compliance?
306. How do you monitor restore job duration for SLA enforcement?
307. How do you detect inconsistent data during restore in multi-region setups?
308. How do you optimize PITR restores for minimal downtime?
309. How do you monitor throughput usage during backup and restore operations?
310. How do you detect and resolve restore errors caused by schema evolution?
311. How do you optimize cross-region restores for global containers?
312. How do you implement automated alerts for backup anomalies?
313. How do you monitor restore success for high-frequency backup schedules?
314. How do you detect failed incremental backups in large-scale systems?
315. How do you implement cross-subscription restore processes?
316. How do you optimize backup performance without affecting production workloads?
317. How do you monitor restore metrics for multi-tenant containers?
318. How do you detect data inconsistencies caused by failed restores?
319. How do you implement automated PITR restore verification for SLAs?
320. How do you optimize restore performance for containers with large JSON documents?

Performance Tuning & Monitoring (321–400)

321. How do you monitor RU/s consumption trends for performance optimization?
322. How do you detect queries causing excessive RU consumption under peak load?
323. How do you optimize queries to avoid cross-partition scans?
324. How do you monitor latency for read-heavy workloads?
325. How do you detect and resolve throttled requests (HTTP 429) in production?
326. How do you optimize batch operations to reduce RU consumption?
327. How do you monitor container-level request charges per operation?
328. How do you detect queries causing uneven partition usage?
329. How do you optimize indexing policies for frequently accessed attributes?

330. How do you monitor server-side stored procedure execution for performance?
331. How do you detect queries that bypass indexing paths and increase RU usage?
332. How do you optimize queries for nested JSON documents?
333. How do you monitor continuation token usage for large result sets?
334. How do you detect inefficient filter or projection operations?
335. How do you optimize cross-partition queries for minimal latency?
336. How do you monitor query execution times in multi-region setups?
337. How do you detect high-latency queries caused by large payloads?
338. How do you optimize stored procedures for high-concurrency workloads?
339. How do you monitor request charges for triggers and user-defined functions?
340. How do you detect queries causing hot partition throttling?
341. How do you optimize analytical queries in analytical store for RU efficiency?
342. How do you monitor throughput utilization for peak-hour workloads?
343. How do you detect inefficient server-side operations affecting performance?
344. How do you optimize queries for frequent update/delete operations?
345. How do you monitor query latency for multi-tenant applications?
346. How do you detect queries causing uneven global partition usage?
347. How do you optimize batch reads/writes for high-throughput workloads?
348. How do you monitor container-level latency under heavy load?
349. How do you detect queries causing replication lag in multi-region setups?
350. How do you optimize queries for high-cardinality attributes?
351. How do you monitor request charges for multi-region writes?
352. How do you detect queries causing excessive cross-region traffic?
353. How do you optimize queries using partition keys and sort keys efficiently?
354. How do you monitor batch query execution for SLA compliance?
355. How do you detect queries that generate large network payloads?
356. How do you optimize queries to minimize cross-partition traffic?
357. How do you monitor change feed processing latency?
358. How do you detect queries causing spikes in server-side CPU usage?
359. How do you optimize queries for hierarchical or nested datasets?
360. How do you monitor analytical store queries for RU consumption?
361. How do you detect cross-region queries affecting performance?
362. How do you optimize queries for containers with frequent writes?
363. How do you monitor request charges for high-volume queries?
364. How do you detect queries that cause hot partition issues?
365. How do you optimize multi-attribute queries for minimal RU consumption?
366. How do you monitor continuation token efficiency for large datasets?
367. How do you detect queries bypassing indexes affecting RU efficiency?
368. How do you optimize queries for time-range and aggregation operations?
369. How do you monitor analytical query performance trends?
370. How do you detect inefficient batch operations affecting throughput?
371. How do you optimize queries to reduce read/write conflicts in multi-master setups?
372. How do you monitor server-side function performance for SLA compliance?

373. How do you detect queries causing uneven partition RU consumption?
374. How do you optimize queries for frequent analytical operations?
375. How do you monitor latency spikes for cross-region queries?
376. How do you detect queries that generate high network traffic?
377. How do you optimize queries for multi-region replicated containers?
378. How do you monitor throughput distribution across partitions?
379. How do you detect conditional write failures under heavy load?
380. How do you optimize stored procedures and triggers for performance?
381. How do you monitor query efficiency for mixed read/write workloads?
382. How do you detect inefficient queries causing hot partitions?
383. How do you optimize queries for large JSON payload attributes?
384. How do you monitor batch insert/update throughput trends?
385. How do you detect queries causing cross-partition latency spikes?
386. How do you optimize queries using selective projections?
387. How do you monitor query execution in multi-tenant setups?
388. How do you detect queries that exceed allocated RU/s?
389. How do you optimize queries for containers with frequent TTL expiration?
390. How do you monitor change feed consumer performance?
391. How do you detect high-latency queries affecting analytical dashboards?
392. How do you optimize queries for global multi-region consistency?
393. How do you monitor query cost trends in high-traffic containers?
394. How do you detect queries causing uneven storage growth across partitions?
395. How do you optimize queries for containers with multiple GSIs?
396. How do you monitor partition-level throughput usage for SLA enforcement?
397. How do you detect queries generating excessive server-side processing?
398. How do you optimize queries for analytical store integration?
399. How do you monitor query efficiency for time-series datasets?
400. How do you detect queries causing hotspots in multi-region writes?

Security, Hot Partitions & Extreme Scenarios (401–500)

401. How do you implement fine-grained role-based access control in Cosmos DB?
402. How do you monitor unauthorized access attempts using Azure Monitor?
403. How do you implement attribute-level encryption for sensitive data?
404. How do you rotate Cosmos DB keys without downtime?
405. How do you detect hot partitions causing SLA breaches?
406. How do you monitor throughput utilization for security-critical containers?
407. How do you optimize partition key selection to avoid hot partitions?
408. How do you detect spikes in write-heavy workloads affecting performance?
409. How do you implement automated failover testing in global setups?
410. How do you monitor endpoint health for multi-region Cosmos DB accounts?
411. How do you detect and mitigate throttling in high-concurrency scenarios?
412. How do you optimize batch operations to handle extreme loads?
413. How do you monitor cross-region replication latency in multi-master setups?

414. How do you detect replication conflicts during global writes?
415. How do you optimize queries for extreme high-throughput workloads?
416. How do you monitor request charges for SLA enforcement?
417. How do you detect queries causing uneven RU consumption across partitions?
418. How do you implement automated alerts for performance anomalies?
419. How do you optimize analytical queries under heavy write traffic?
420. How do you monitor latency for change feed consumers under extreme loads?
421. How do you detect inefficient batch writes causing throttling?
422. How do you optimize cross-partition queries for minimal latency?
423. How do you monitor throughput distribution in multi-tenant environments?
424. How do you detect conditional write conflicts affecting high-volume containers?
425. How do you optimize queries for frequent TTL-based deletes?
426. How do you monitor global write consistency in multi-region deployments?
427. How do you detect queries causing replication lag under peak load?
428. How do you optimize stored procedures for extreme concurrency?
429. How do you monitor query execution cost for multi-region queries?
430. How do you detect queries generating hot partition throttling?
431. How do you optimize queries for containers with large JSON documents?
432. How do you monitor request charges for triggers and UDFs under load?
433. How do you detect queries causing cross-region network congestion?
434. How do you implement partition key strategies for multi-master writes?
435. How do you monitor global container performance trends?
436. How do you detect inefficient indexing paths affecting multi-region queries?
437. How do you optimize queries for analytical store operations?
438. How do you monitor latency for cross-partition operations?
439. How do you detect queries causing uneven storage and RU distribution?
440. How do you optimize queries for high-frequency insert/update workloads?
441. How do you monitor request charges for multi-attribute queries?
442. How do you detect queries causing performance degradation in production?
443. How do you optimize queries for hierarchical or nested datasets?
444. How do you monitor batch operation throughput in high-volume containers?
445. How do you detect cross-region write conflicts during peak hours?
446. How do you optimize queries using continuation tokens efficiently?
447. How do you monitor container-level latency for SLA compliance?
448. How do you detect queries generating excessive server-side computation?
449. How do you optimize change feed processing for extreme write workloads?
450. How do you monitor multi-region analytical query performance?
451. How do you detect and mitigate hot partition issues dynamically?
452. How do you optimize queries for time-series data with high ingest rates?
453. How do you monitor global write latency across regions?
454. How do you detect inefficient queries causing RU spikes in production?
455. How do you optimize queries for large-scale multi-tenant environments?
456. How do you monitor analytical store queries for high-frequency access?

457. How do you detect queries causing uneven partition key utilization?
458. How do you optimize queries for low-latency multi-region reads?
459. How do you monitor throughput utilization in extreme high-load scenarios?
460. How do you detect cross-region replication bottlenecks in real-time?
461. How do you optimize batch writes for high-concurrency workloads?
462. How do you monitor container-level RU/s for SLA adherence?
463. How do you detect queries bypassing indexes affecting global performance?
464. How do you optimize queries for multi-region read/write patterns?
465. How do you monitor change feed latency under peak traffic?
466. How do you detect queries causing hot partition throttling?
467. How do you optimize queries for selective projections to reduce RU usage?
468. How do you monitor global throughput distribution across regions?
469. How do you detect conditional write conflicts under extreme load?
470. How do you optimize analytical queries for containers with high write volume?
471. How do you monitor cross-partition query latency for multi-tenant workloads?
472. How do you detect inefficient server-side logic affecting performance?
473. How do you optimize queries for frequent TTL expiration scenarios?
474. How do you monitor request charges per stored procedure execution?
475. How do you detect queries causing uneven GSI usage?
476. How do you optimize queries for multi-region analytics?
477. How do you monitor endpoint-level latency for global applications?
478. How do you detect queries causing excessive RU/s consumption spikes?
479. How do you optimize batch operations under multi-region replication?
480. How do you monitor change feed consumer health under high-load scenarios?
481. How do you detect queries causing storage hotspots in global containers?
482. How do you optimize queries for hierarchical data in multi-master setups?
483. How do you monitor global write availability during peak traffic?
484. How do you detect queries causing performance degradation in multi-tenant workloads?
485. How do you optimize queries for large JSON payloads to reduce RU consumption?
486. How do you monitor analytical query throughput in real-time?
487. How do you detect cross-region query latency affecting SLAs?
488. How do you optimize queries for time-range and aggregation operations?
489. How do you monitor request charges for extreme high-concurrency workloads?
490. How do you detect queries causing uneven partition usage during peak hours?
491. How do you optimize stored procedures for global multi-region applications?
492. How do you monitor throughput utilization in multi-master write setups?
493. How do you detect queries generating high cross-region network traffic?
494. How do you optimize change feed processing for large-scale workloads?
495. How do you monitor RU/s consumption per operation in extreme scenarios?
496. How do you detect hot partitions affecting multi-region consistency?
497. How do you optimize queries for analytical workloads on heavily updated containers?
498. How do you monitor container-level latency and throughput for SLA compliance?
499. How do you detect cross-region replication conflicts in real-time?

500. How do you implement extreme performance tuning for Cosmos DB while ensuring high availability, minimal latency, and cost efficiency?

All 500 Azure Cosmos DB real-time scenario-based interview questions, covers:

- Setup & Configuration
- Query Performance & Index Optimization
- Multi-Region Replication & Global Distribution
- Backup & Recovery
- Performance Tuning & Monitoring
- Security, Hot Partitions & Extreme Scenarios

These serve a **complete guide for interview preparation, real-time scenario discussions, or DBA skill assessment.**

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